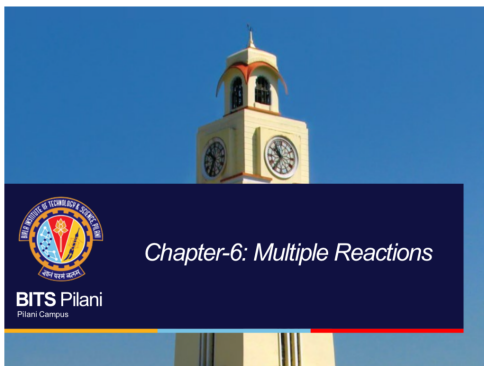




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CHE F311 Kinetics & Reactor Design



## Objectives

- Define different types of **selectivity** and **yield**
- Choose a reaction system that would **maximize the selectivity of the desired product** given the rate laws for all the reactions occurring in the system.
- Describe the algorithm used to design reactors with multiple reactions.
- **Size reactors to maximize the selectivity** and to determine the species concentrations in a batch reactor, semi-batch reactor, CSTR, PFR, and PBR, systems.

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CHE F311 Kinetics & Reactor Design



## Introduction

- Multiple reactions
- Desired and **undesired** reactions
- Key challenge: minimize the undesired reactions
  - Optimize the batch time, space time, operating parameters

### Rules of thumb

- For reactions in parallel, the concentration level of reactants is the key to proper control of product distribution.
  - A **high reactant concentration** favors the reaction of **higher order**, a **low concentration** favors the reaction of **lower order**,
  - Concentration level has no effect on the product distribution for reactions of the same order

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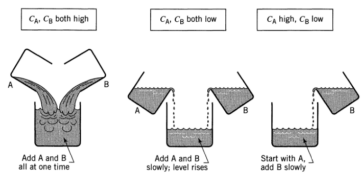
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# Patterns



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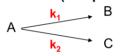
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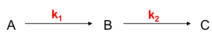


# Definition of Multiple Reaction

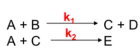
## Parallel rxns (competing rxns)



## Series rxns (consecutive rxns)



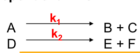
## Complex rxns (Parallel + Series rxns)



Denbigh network:



## Independent rxns



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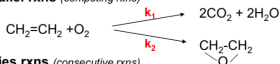
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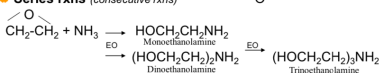


## Examples

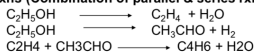
### ⚙ Parallel rxns (competing rxns)



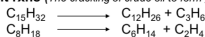
Series rxns (consecutive rxns)



- Complex Rxns (Combination of parallel & series rxns)



- **Independent rxns** (The cracking of crude oil to form gasoline)



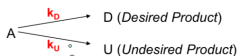
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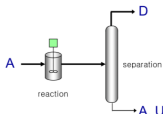
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## Parallel Reactions



**Competing or side rxn**



Rxn-separation system producing both D & U

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### The economic incentive

Maximize the formation of D  
Minimize the formation of U



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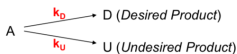
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## Rate selectivity parameter, $S_{DU}$



The rate laws are

Tells us how one product is favored over another when we have multiple reactions

$$-r_A = k_D C_A^{\alpha_1} + k_U C_A^{\alpha_2}$$

$$r_D = k_D C_A^{\alpha_1}$$

$$r_U = k_U C_A^{\alpha_2}$$

We want to maximize  $S_{DU}$

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_D}{k_U} C_A^{\alpha_1 - \alpha_2} = \frac{\text{Rate of formation of D}}{\text{Rate of formation of U}}$$

**Rate selectivity parameter = Instantaneous selectivity**

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## Overall Selectivity, $S_{DU}$

### Overall Selectivity

$$\tilde{S}_{DU} = \frac{F_D}{F_U} = \frac{\text{Exit molar flow rate of desired product}}{\text{Exit molar flow rate of undesired product}}$$

For Batch Reactor

$$\tilde{S}_{DU} = \frac{N_D}{N_U}$$

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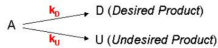
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## Comparison between SDU and $\tilde{S}_{DU}$ in a CSTR

$S_{DU}$  and  $\tilde{S}_{DU}$  in a CSTR



$$S_{DU} = \frac{r_D}{r_U} \quad \text{and} \quad \tilde{S}_{DU} = \frac{F_D}{F_U}$$

$$F_D = r_D V \quad \text{and} \quad F_U = r_U V$$

$$\tilde{S}_{DU} = \frac{F_D}{F_U} = \frac{r_D V}{r_U V} = \frac{r_D}{r_U} = S_{DU}$$



## Instantaneous Yield and Overall Yield

### Instantaneous Yield (Basis: Reaction Rate)

$$Y_D = \frac{r_D}{-r_A}$$

### Overall Yield (Basis: Molar Flow Rate)

For a batch system:  $\tilde{Y}_D = \frac{N_D}{N_{A0} - N_A} = \frac{\text{Mole of desired product formed at the end of reaction}}{\text{Number of moles of key reactant consumed}}$

For a flow system:  $\tilde{Y}_D = \frac{F_D}{F_{A0} - F_A}$



## Overall Selectivity

$$\tilde{S}_{DU, \text{flow system}} = \frac{F_D}{F_U} = \frac{\text{exit molar flow rate of D}}{\text{exit molar flow rate of U}}$$

$$\tilde{S}_{DU, \text{batch system}} = \frac{N_D}{N_U} = \frac{\text{the number of moles of D at the end of the rxn time}}{\text{the number of moles of U at the end of the rxn time}}$$

## Overall Yield

$$\tilde{Y}_{D, \text{batch system}} = \frac{N_D}{N_{A0} - N_A} = \frac{\text{the number of moles of D at the end of the rxn time}}{\text{the number of moles of the key reactant, A, that have been consumed}}$$

$$\tilde{Y}_{D, \text{flow system}} = \frac{F_D}{F_{A0} - F_A} = \frac{\text{the number of moles of D at the exit}}{\text{the number of moles of A that have been consumed}}$$

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## Important

Different definitions for selectivity and yield



Check carefully to ascertain the definition intended by the author

From an *economic standpoint*,  
overall selectivities and yields are important  
in determining profits

The instantaneous selectivities give insights  
in choosing reactors and reaction schemes  
that will help maximize the profit

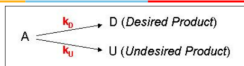
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## Maximizing $S_{DU}$ in Parallel Reactions



Case 1:  $\alpha_1 > \alpha_2$ ,  $a = \alpha_1 - \alpha_2$

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_D}{k_U} C_A^a$$

maximize  $S_{DU}$

- keeping the concentration of reactant A as high as possible during the rxn
- in gas phase rxn, we should run it without inerts and at high pressures to keep  $C_A$  high
- in liquid phase rxn, the use of diluents should be kept to a minimum
- use a batch or plug-flow reactor



## Maximizing $S_{DU}$ for one reactant

Case 2:  $\alpha_2 > \alpha_1$ ,  $a = \alpha_2 - \alpha_1$

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_D}{k_U} \frac{1}{C_A^a}$$

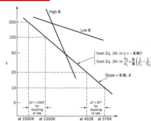
maximize  $S_{DU}$

- keeping the concentration of reactant A as low as possible during the rxn
- in gas phase rxn, we should run it with inerts and at low pressures to keep  $C_A$  low
- in liquid phase rxn, the use of diluents should be kept to a maximum
- use a CSTR or recycle reactor (product stream act as a diluent)

## Maximizing $S_{DU}$ for one reactant



$$\frac{k_D}{k_U} = \frac{A_D}{A_U} e^{-\frac{E_D - E_U}{RT}}$$

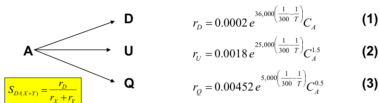


### Case 3: $E_D > E_U$

- $k_D$  ( $r_D$ ) increases more rapidly with increasing temperature than does the  $k_U$ .
- keeping the temperature as high as possible to maximize  $S_{DU}$ .

### Case 4: $E_U > E_D$

- keeping the temperature as low as possible to maximize  $S_{DU}$
- not so low that the desired rxn does not proceed to any significant extent.



- Pre-exponential factors are comparable
- the activation energies of (1) & (2) are much greater than that of (3).
- at high temperature  $r_Q \ll r_D, r_U$

$$S_{DQ} = \frac{r_D}{r_Q} \cong \text{very large} \quad (4)$$

$$S_{DU} = \frac{r_D}{r_U} = \frac{0.11 e^{11,000 \left( \frac{1}{300} - \frac{1}{T} \right)}}{C_A^{0.5}} \quad (5)$$

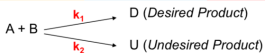
- From (5),  $S_{DU}$  is minimized at low concentration of A



1. High Temperatures (to minimize the formation of Q)
2. Low concentration of A (to minimize the formation of U)
  - Adding inerts, using low pressures, using CSTR or recycled reactor



## Reactor Selection and Operating Conditions



The rate laws are

$$-r_A = k_1 C_A^{\alpha_1} C_B^{\beta_1} + k_2 C_A^{\alpha_2} C_B^{\beta_2}$$

$$r_D = k_1 C_A^{\alpha_1} C_B^{\beta_1}$$

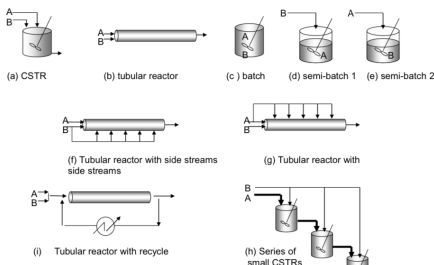
$$r_U = k_2 C_A^{\alpha_2} C_B^{\beta_2}$$

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_1}{k_2} C_A^{\alpha_1 - \alpha_2} C_B^{\beta_1 - \beta_2}$$

Rate selectivity parameter=Instantaneous selectivity



## Schemes for maximizing the desired product



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## Example



For the parallel reaction



$$r_D = k_1 C_A^{\alpha_1} C_B^{\beta_1}$$

$$r_U = k_2 C_A^{\alpha_2} C_B^{\beta_2}$$

**Case I:**  $\alpha_1 > \alpha_2$ ,  $\beta_1 > \beta_2$ ,  $a = \alpha_1 - \alpha_2 > 0$ ,  $b = \beta_1 - \beta_2 > 0$ 

the rate selectivity parameter

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_1}{k_2} C_A^a C_B^b$$

To maximize the  $S_{DU}$ , maintain the concentration of both A and B as high as possible

- ✳ a tubular reactor (b)
- ✳ a batch reactor (c)
- ✳ high pressures (if gas phase), reduce inert

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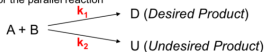
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## Example

for the parallel reaction



**Case II** :  $\alpha_1 > \alpha_2$ ,  $\beta_1 < \beta_2$ ,  $a = \alpha_1 - \alpha_2 > 0$ ,  $b = \beta_2 - \beta_1 > 0$

the rate selectivity parameter

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_1 C_A^a}{k_2 C_B^b}$$

To maximize the  $S_{DU}$ , maintain  $C_A$  high and  $C_B$  low.

- ✳ a semibatch reactor in which B is fed slowly into A. (d)
- ✳ a tubular reactor with side stream of B continually (f)
- ✳ a series of small CSTRs with A fed only to the first reactor (h)

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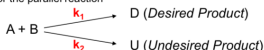
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## Example

for the parallel reaction



**Case III** :  $\alpha_1 < \alpha_2$ ,  $\beta_1 < \beta_2$ ,  $a = \alpha_2 - \alpha_1 > 0$ ,  $b = \beta_2 - \beta_1 > 0$

the rate selectivity parameter

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_1}{k_2 C_A^a C_B^b}$$

To maximize the  $S_{DU}$ , maintain the concentration of both A and B as low as possible

- ✳ a CSTR (a)
- ✳ a tubular reactor in which there is a large recycle ratio (i)
- ✳ a feed diluted with inert material
- ✳ low pressures (if gas phase)

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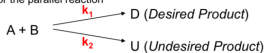
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## Example

for the parallel reaction



**Case IV:**  $\alpha_1 < \alpha_2$ ,  $\beta_1 > \beta_2$ ,  $a = \alpha_2 - \alpha_1 > 0$ ,  $b = \beta_1 - \beta_2 > 0$

the rate selectivity parameter

$$S_{DU} = \frac{r_D}{r_U} = \frac{k_1 C_B^b}{k_2 C_A^a}$$

To maximize the  $S_{DU}$ , maintain the concentration of both A and B as high as possible

- ✚ a semibatch reactor in which A is slowly fed to B (Figure 6.3(e))
- ✚ a tubular reactor with side stream of A (Figure 6.3(g))
- ✚ a series of small CSTRs with fresh B fed to each reactor (Figure 6.3(h))

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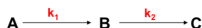
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## Maximizing the desired product in series reaction



✚ In **parallel** rxns, maximize the desired product

- ① by adjusting the reaction conditions
- ① by choosing the proper reactor

✚ In **series** rxns, maximize the desired product

- ① by adjusting the space-time for a flow reactor
- ① by choosing real-time for a batch reactor

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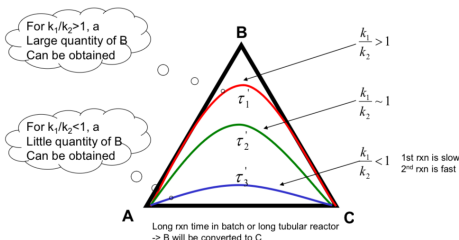
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## Series Reaction



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## Quantitative Treatment of Product Distribution and of Reactor Size

- If rate equations are known for the individual reactions, we can quantitatively determine product distribution and reactor-size requirements

- Instantaneous fractional yield of R

$$\varphi = \left( \frac{\text{moles R formed}}{\text{moles A reacted}} \right) = \frac{dC_R}{-dC_A}$$

- Overall yield

$$\Phi = \left( \frac{\text{all R formed}}{\text{all A reacted}} \right) = \frac{C_{R_f}}{C_{A0} - C_{A_f}} = \frac{C_{R_f}}{(-\Delta C_A)} = \bar{\varphi}_{\text{in reactor}}$$

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$$\text{For PFR: } \Phi_p = \frac{-1}{C_{A0} - C_{Af}} \int_{C_{A0}}^{C_{Af}} \varphi dC_A = \frac{1}{\Delta C_A} \int_{C_{A0}}^{C_{Af}} \varphi dC_A$$

$$\text{For MFR: } \Phi_m = \varphi_{\text{evaluated at } C_{Af}}$$

$$\Phi_m = \left( \frac{d\Phi_p}{dC_A} \right)_{\text{at } C_{Af}} \quad \text{and} \quad \Phi_p = \frac{1}{\Delta C_A} \int_{C_{A0}}^{C_{Af}} \Phi_m dC_A$$

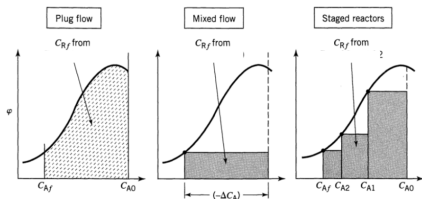


## MFRs series

$$\varphi_1(C_{A0} - C_{A1}) + \dots + \varphi_N(C_{A,N-1} - C_{AN}) = \Phi_{N \text{ mixed}} (C_{A0} - C_{AN})$$

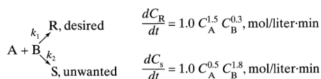
$$\Phi_{N \text{ mixed}} = \frac{\varphi_1(C_{A0} - C_{A1}) + \varphi_2(C_{A1} - C_{A2}) + \dots + \varphi_N(C_{A,N-1} - C_{AN})}{C_{A0} - C_{AN}}$$

$$C_{Rf} = \Phi(C_{A0} - C_{Af})$$



## Product distribution

Consider the aqueous reactions



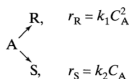
For 90% conversion of A find the concentration of R in the product stream. Equal volumetric flow rates of the A and of B streams are fed to the reactor, and each stream has a concentration of 20 mol/liter of reactant.

The flow in the reactor follows.

- (a) Plug flow
- (b) Mixed flow
- (c) Plug flow with side entrances of B



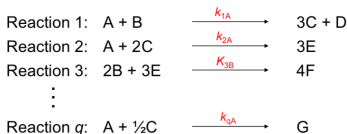
Substance A in the liquid phase produces R and S by the following reactions:



The feed ( $C_{\text{A}0} = 1.0$ ,  $C_{\text{R}0} = 0$ ,  $C_{\text{S}0} = 0.3$ ) enters two mixed flow reactors in series ( $\tau_1 = 2.5$  min,  $\tau_2 = 10$  min). Knowing the composition in the first reactor ( $C_{\text{A}1} = 0.4$ ,  $C_{\text{R}1} = 0.2$ ,  $C_{\text{S}1} = 0.7$ ), find the composition leaving the second reactor.



## Net Rates of Reaction



Net rates of reaction of A and B:

$$r_{\text{A}} = r_{1\text{A}} + r_{2\text{A}} + r_{3\text{A}} + \cdots + r_{q\text{A}} = \sum_{i=1}^q r_{i\text{A}}$$

$$r_{\text{B}} = r_{1\text{B}} + r_{2\text{B}} + r_{3\text{B}} + \cdots + r_{q\text{B}} = \sum_{i=1}^q r_{i\text{B}}$$



### Total rate of formation of each species from all reaction

| Reaction Number | Reaction Stoichiometry                                            |
|-----------------|-------------------------------------------------------------------|
| 1               | $\nu_{11}A_1 + \nu_{12}A_2 \rightarrow \nu_{1j}A_j$               |
| 2               | $\nu_{21}A_1 + \nu_{2j}A_j \rightarrow \nu_{23}A_3 + \nu_{24}A_4$ |
| .               | .                                                                 |
| .               | .                                                                 |
| $i$             | $\nu_{ij}A_j + \nu_{ik}A_k \rightarrow \nu_{ip}A_p$               |
| .               | .                                                                 |
| .               | .                                                                 |
| $q$             | $\nu_{qj}A_j + \nu_{q3}A_3 \rightarrow \nu_{q5}A_5$               |

The net rate of rxn for A:

$$r_j = \sum_{i=1}^q r_{ij}$$

$q$  rxns taking place  
 $r_{ij}$  → species  
 $i$  → reaction number

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